

Jithesh Singamsetty

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4 years shipping production **LLM** and **Generative AI** systems at **Qualcomm**, **JPMorgan Chase**, and **HCL Technologies** across semiconductor, financial services, and enterprise domains. Specialized in **LLM fine-tuning (QLoRA, LoRA, RLHF, DPO)**, **RAG** pipelines, and **Agentic AI** workflows using **LangChain**, **LlamaIndex**, and **Pinecone**. Proven track record compressing 7B-parameter models to sub-1GB footprints via **INT8/INT4 quantization** and **structured pruning**, cutting inference latency from 180ms to 58ms on **Snapdragon** edge devices, and sustaining 2,400 concurrent requests/sec via **Triton Inference Server**. Deep expertise in **PyTorch**, **HuggingFace Transformers**, **AWS SageMaker**, **GCP Vertex AI**, and end-to-end **MLOps** with **Docker**, **Kubernetes**, and **MLflow**. Delivered \$2.1M+ in measurable business impact across fraud detection, demand forecasting, and enterprise NLP.

PROFESSIONAL EXPERIENCE

AI/ML Engineer @ [Qualcomm, USA](#)

Dec 2025 – Present

- Architected on-device neural network inference pipelines using **Qualcomm AI Engine SDK**, cutting end-to-end model latency from 180ms to **58ms** on Snapdragon-powered edge devices.
- Fine-tuned transformer-based LLMs (7B parameters) for on-device NLU tasks using **QLoRA**, achieving sub-1GB model footprints without measurable accuracy degradation on benchmark suites.
- Applied **INT8 quantization** and **structured pruning** to compress vision models by **62%**, enabling deployment across 4 mobile product lines previously constrained by memory budgets.
- Deployed optimized models via **Triton Inference Server**, sustaining 2,400 concurrent inference requests per second with P99 latency under 95ms.
- Established **MLOps CI/CD pipelines** with **Docker**, **Kubernetes**, and **MLflow**, **Weights & Biases** (W&B), shrinking model release cycles from 3 weeks to 5 days across 6 engineering teams.
- Co-designed NPU scheduling workflows with hardware architects, raising AI accelerator utilization from 54% to **87%** across production chipsets.

AI/ML Engineer @ [HCL Technologies, USA](#)

Feb 2025 – Dec 2025

- Engineered an end-to-end **NLP** document classification pipeline processing 500K+ enterprise documents monthly, reaching **94.3% accuracy** across 18 document categories.
- Fine-tuned **RoBERTa-large** on 1.2M domain-specific tokens, lifting named entity recognition **F1 from 71 to 89** an 18-point gain over the pre-trained baseline.
- Implemented **RLHF and DPO**-aligned text summarization for legal contracts using human preference data from 3,200 annotated examples, cutting manual review time by 4 hours per analyst per week.
- Deployed 9 production inference endpoints on **AWS SageMaker** with auto-scaling, maintaining 99.6% uptime SLA across 12 enterprise client environments.
- Automated feature engineering workflows with **Apache Spark** across a 40-node cluster, reducing per-batch preprocessing runtime from 6 hours to **38 minutes**.
- Introduced model drift monitoring via **Evidently AI**, catching 3 silent model degradations before they impacted client-facing SLAs.

AI/ML Engineer Intern @ [JPMorgan Chase & Co., USA](#)

Aug 2024 – Nov 2024

- Constructed a real-time fraud detection system using **gradient boosting ensembles** trained on 10M+ transaction records, achieving **91.4% precision** and **88.7% recall** on held-out test data.
- Developed **RAG-based earnings** call analysis tool using **LangChain**, **LlamaIndex**, and **OpenAI API** (GPT-4), with Pinecone as the vector store, surfacing actionable investment signals adopted into 2 active research workflows.
- Transformed 10M+ financial transaction records using **PySpark** and Pandas, producing feature sets that reduced model training time from 9 hours to **2.5 hours**.
- Wired real-time model monitoring dashboards with **Grafana** and **Prometheus**, enabling drift detection within 12 minutes of threshold breach down from 45-minute manual checks.
- Developed **Agentic AI-based** inventory replenishment system using **LangChain** tool-calling workflows, automating purchase order generation for 8 distribution centers and reducing procurement cycle time from **5 days to 1 day**.

Machine Learning Engineer @ [Redington, India](#)

Nov 2020 – Dec 2022

- Built **LSTM** and **ARIMA** demand forecasting models across 8 regional distribution centers, lowering overstock inventory value by **\$2.1M** in the first operating year.
- Designed a **CNN-based** visual quality inspection system that automated defect detection for 3 product lines, replacing 100% of manual sampling review for those categories.
- Served production models via **vLLM-compatible** REST APIs, supporting 1,100 daily inference requests across 5 internal business applications with 99.2% uptime.

- Consolidated 14 siloed training jobs into unified **GCP Vertex AI** pipelines, cutting monthly compute spend from \$18,400 to **\$9,700**.
- Sustained model performance above agreed SLA thresholds for **18 consecutive months** through quarterly retraining and automated evaluation harnesses.
- Guided 2 junior analysts on feature engineering and evaluation best practices, halving their onboarding timeline from 8 weeks to **4 weeks**.
- Evaluated and benchmarked 6 classification algorithms for customer churn prediction, selecting **XGBoost** as the final model with **88% AUC** directly informing a retention campaign strategy.
- Cleaned and normalized 220K+ customer records using **Python** and **SQL**, producing reusable preprocessing scripts the data team adopted as the team standard.
- Visualized model evaluation metrics and business KPIs in **Tableau**, enabling non-technical stakeholders to self-serve insights without analyst support.

PROJECTS

EdgeLM - On-Device LLM Compression Toolkit – PyTorch, QLoRA, ONNX Runtime, TensorRT, Triton, Weights & Biases

- Compressed a 7B-parameter LLaMA-2 model to 940MB using **QLoRA + INT4 quantization + magnitude pruning**, retaining **97.1% MMLU score**, validated on Snapdragon 8 Gen 3 hardware.
- Exported final checkpoints to **ONNX** and **TFLite**, reducing cross-platform integration effort from 3 days to under 4 hours per target device.

ClauseIQ - Legal Document NER & Clause Classifier – RoBERTa, FastAPI, PostgreSQL, Docker

- Fine-tuned **RoBERTa-base** on 14,000 annotated legal clauses across 6 contract types, hitting **91.6% macro-F1** outperforming a GPT-3.5 zero-shot baseline by 11 points.
- Packaged the model as a **FastAPI** service with async batch endpoints, processing 800-page contracts in under 9 seconds on a single t3.medium instance.

TxShield - Streaming Transaction Anomaly Detector – XGBoost, Kafka, Redis, Grafana, PySpark, Pinecone

- Trained an **XGBoost ensemble** on 22M transactions with **SMOTE oversampling**, reaching **93.4% F1** on the minority fraud class at 40ms p95 scoring latency via **Redis feature cache**.
- Streamed live transaction events through **Kafka** and surfaced model confidence scores in a **Grafana** dashboard, cutting mean alert response time from 38 minutes to under **3 minutes**.

StockSense - Hybrid Demand Forecasting Engine – LSTM, Prophet, GCP Vertex AI, BigQuery

- Combined **Facebook Prophet** for trend decomposition with an **LSTM residual network** across 220 SKUs, achieving **11.4% MAPE**, a 6.2-point reduction versus the standalone ARIMA baseline.
- Scheduled weekly retraining on **GCP Vertex AI Pipelines** with **BigQuery** as the feature store, giving procurement teams self-serve forecast access through a Looker Studio dashboard.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, Scala

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, HuggingFace Transformers

LLM & Generative AI: LLM Fine-tuning (QLoRA, LoRA, DPO, PPO), RLHF, RAG, Prompt Engineering, LangChain, LlamaIndex, vLLM, Triton Inference Server, TensorRT, ONNX Runtime, CUDA, OpenAI API

Vector Databases & Agents: Pinecone, Weaviate, Qdrant, pgvector, LangGraph, CrewAI, Agentic AI

AI Domains: NLP, Computer Vision, Anomaly Detection, Time-Series Forecasting, Model Compression (Quantization, Pruning), Foundation Models (LLaMA-2, Mistral, GPT-4), Agentic Systems

MLOps & Cloud: AWS SageMaker, GCP Vertex AI, Azure ML (AzureML SDK), Docker, Kubernetes, MLflow, Kubeflow, Airflow, CI/CD, Evidently AI, Weights & Biases (W&B), DVC

Data Engineering: Apache Spark, PySpark, Kafka, Pandas, NumPy, PostgreSQL, MongoDB, Snowflake

Tools: Git, Jupyter, VS Code, Grafana, Prometheus, Tableau, JIRA

EDUCATION

University of North Texas

TX, USA

Masters in Computer Science and Engineering

Jan 2023 - Dec 2024

Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Statistical Modeling, Data Mining, Natural Language Processing (NLP), Computer Vision, and Big Data Analytics.

CERTIFICATIONS

- E-Box: Python Developer